1.4 States of Matter

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.4 States of Matter
Difficulty	Hard

Time allowed: 80

Score: /63

Percentage: /100

Question 1a

Phosphine, PH_3, is a gas formed by heating phosphorous acid, H_3PO_3, in the absence of air.

$$4H_3PO_3\ (s) \rightarrow PH_3\ (g) + 3H_3PO_4\ (s)$$

State the shape and bond angle in PH_3 (g).

[1 mark]

Question 1b

3.45×10^{-2} mol of H_3PO_3 is completely decomposed at 100 kPa pressure and 210 °C.

Calculate the volume occupied, in cm^3, by the phosphine gas produced.

[3 marks]

Question 1c

1.85 g of white phosphorus was reacted with 75.00 cm^3 of 1.25 mol dm^{-3} sodium hydroxide solution to make phosphine.

$$P_4\ (s) + 3OH^-\ (aq) + 3H_2O\ (l) \rightarrow PH_3\ (g) + 3H_2PO_2^-\ (aq)$$

Deduce which chemical is the limiting reagent.

[3 marks]

Question 1d

Use the information in part (c) to calculate the volume, in cm^3, of phosphine that was produced at room temperature and pressure. Give your answer to three significant figures.

[1 mark]

Question 2a

Oxygen exists as a diatomic gas, O_2 (g). A sample of O_2 (g) was made during a chemical reaction. When measured at 303 kPa and 28 °C the sample occupied a volume of 95.0 cm^3.

Calculate the mass of oxygen formed.

For this calculation, assume that oxygen behaves as an ideal gas under these conditions.

[2 marks]

Question 2b

Calculate the number of electrons involved in the bonding of this sample of oxygen. You may find it helpful to use your answer to **(a)**.

[2 marks]

Question 2c

O_2 (g) does not behave as an ideal gas under these conditions.

Explain why O_2 (g) behaves even **less** ideally at:

- very high pressures
- very low temperatures

[2 marks]

Question 2d

The homologous series of alkanes undergo combustion with oxygen.

A 2.0 dm^3 flask contains 10.84 g of a gaseous alkane, **X**. The pressure in the flask is 300 kPa and the temperature is 20 ^oC.

Write an equation for the complete combustion of **X**.

[1 mark]

Question 3a

Airbags are safety devices fitted to modern cars. They are designed to rapidly inflate in the event of a collision, in order to protect the occupants of the car from the effects of the impact, and then quickly deflate.

The inflation of an airbag depends on the chemical decomposition of sodium azide, NaN_3.

The azide ion, N_3^-, contains one triple bond. Draw a 'dot-and-cross' diagram to show the arrangement of outer electrons present in an azide ion.

[3 marks]

Question 3b

Suggest **three** properties of sodium azide. Explain your answer.

[3 marks]

Question 3c

550 g of sodium azide, sodium hydroxide and ammonia were produced during the chemical reaction of sodium amide, $NaNH_2$, and dinitrogen monoxide. The yield of sodium azide in this reaction was 95.0%.

Calculate the mass, in grams, of sodium amide required for this reaction.

[5 marks]

Question 3d

In a serious collision, the airbag deploys and rapidly fills with nitrogen as sodium azide decomposes.

$$2NaN_3\ (s) \rightarrow 2Na\ (s) + 3N_2\ (g)$$

Calculate the mass of sodium azide that must decompose in order to inflate an airbag to a volume of 7.50×10^{-2} m^3 at a pressure of 150 kPa and temperature of 35 °C.

[4 marks]

Question 3e

Sodium azide is toxic. It can be destroyed by reacting it with acidified nitrous acid, HNO_2.

i)

Balance the chemical equation for the reaction of sodium azide with acidified nitrous acid

$$2NaN_3 + ...HNO_2 + ...HCl \rightarrow ...N_2 + ...NO + ...NaCl + 2H_2O$$

[2]

ii)

Calculate the volume of gas released, in dm^3, when 250 cm^3 volume of nitrous acid is used to destroy 75.0 g of sodium azide at room temperature and pressure.

[3]

[5 marks]

Question 4a

This question is about the melting points of different chemicals.

The melting and boiling points of calcium and chlorine are given in Table 4.1.

Table 4.1

Element	**Melting point / K**	**Boiling point / K**
Chlorine	172	239
Calcium	1115	1757

i)

In terms of structure and bonding, compare the boiling points of chlorine and calcium.

[3]

ii)

Suggest why calcium is a liquid over a much greater temperature range than chlorine.

[1]

[4 marks]

Question 4b

Sulfur dioxide and magnesium oxide are both oxides. The melting point of sulfur dioxide is –72 ^{o}C and the melting point of magnesium oxide is 2852 ^{o}C

i)
Explain, with reference to structure and bonding, the difference in the melting points of sulfur dioxide and magnesium oxide.

[3]

ii)
Ammonia has the same crystalline structure as sulfur dioxide but its melting point is 2 ^{o}C.

Explain the difference in melting point between sulfur dioxide and ammonia.

[2]

[5 marks]

Question 4c

Silver chloride is used in photography films as well as having antiseptic properties.

i)

Silver chloride has a structure similar to that of sodium chloride. Draw a diagram of the silver chloride lattice showing the arrangement of the ions.

[2]

ii)

Explain why silver chloride has a relatively high melting point. You should refer to the difference between the electronegativities of the elements in your answer.

[3]

[5 marks]

Question 4d

Diamond and ice are types of crystals.

Describe the structure and bonding present in both diamond and ice. Explain why the melting point of these two substances is very different. You should refer to the type of bonding present in both diamond and ice in your answer.

[1 mark]

Question 5a

Mercury is a naturally occurring element that is a liquid at room temperature and pressure. Some of its physical properties are shown in Table 5.1.

Table 5.1

	Appearance at room temperature	**Melting point / ^{o}C**	**Boiling point / ^{o}C**	**Electrical conductivity**
Mercury	shiny silver liquid	-38.8	356.7	good

Mercury can react with bromine to form mercury(II) bromide, $HgBr_2$.

Compare and explain the electrical conductivity of mercury and mercury(II) bromide.

[3 marks]

Question 5b

The strength of ionic bonding in different compounds can be compared by using the amount of energy required to separate the ions, as shown in Table 5.2.

Table 5.2

Cation	**Amount of energy required to separate the ions / kJ mol-1**
LiF	1031
NaF	787
MgF2	2962

Using the data from Table 5.2, explain how changes in the cation affect the bond strength in an ionic compound.

[2 marks]

Question 5c

Sodium chloride and iodine are both solids. Sodium chloride does not melt until it reaches a temperature of 1074 K yet iodine sublimes when heated gently, giving off purple vapours. Sodium chloride will conduct electricity when molten and iodine is a very poor conductor of electricity.

i)

State the type of crystal structure of iodine and sodium chloride.

[2]

ii)

Explain why iodine vaporises easily.

[2]

iii)

Explain the differences in electrical conductivity of sodium chloride and iodine.

[3]

[7 marks]

Question 5d

Graphene, graphite and diamond are all forms of solid carbon.

Explain, in terms of structure and bonding, why graphene and graphite are good electrical conductors but diamond is a poor electrical conductor.

[1 mark]

Model Answers: Hard

1a

a) The molecular shape and bond angle of PH_3 is:

- Pyramidal
  AND
  107 °; [1 mark]

[Total; 1 mark]

- Phosphine has 3 N-H bonds, which means that there are 3 bonding pairs
- It is not an electron-deficient compound so it must have a lone / non-bonding pair of electrons to satisfy the octet rule for the phosphorous atom
- So, there are three bonding pairs and 1 lone pair in the structure
- This means that the shape is pyramidal with a corresponding bond angle of 107 °

1b

b) To calculate the volume, in cm^3, of phosphine gas:

- Moles of PH_3 (nPH_3) = 8.625 x 10^{-3}; [1 mark]
- T = 483 K
  AND
  P = 100 000 Pa; [1 mark]
- $V = \dfrac{8.63 \times 10^{-3} \text{ mol} \times 8.31 \text{ J K}^{-1}\text{mol}^{-1} \times 483 \text{ K}}{100\ 000 \text{ Pa}}$ = 3.46 x 10^{-4} (m^3)
  AND
  V = 346 (cm^3) ; [1 mark]

[Total: 3 marks]

- For this ideal gas question, you need to:
  - Convert your units for temperature and pressure, make sure you know these from memory:

    - P = needed in Pa (given as kPa in this question so multiply by 1000 to convert to Pa)
      100 kPa x 1000 = 100000 Pa
    - T = needed in K (given as °C in this question so +273 to convert to K)
      210 ^oC + 273 = 483 K
  - Use the molar ratio from the equation in part (a)
    - $4H_3PO_3$ (s) → PH_3 (g) + $3H_3PO_4$ (s)
    - 4 moles of phosphorous acid produce 1 mole of phosphine
    - So, the number of moles of phosphine = $\frac{3.45 \times 10^{-2}}{4}$ = 8.625 x 10^{-3}
  - Rearrange the $PV = nRT$ equation for volume
    - $V = \frac{nRT}{P}$
  - Insert the values and evaluate the expression
    - $V = \frac{8.63 \times 10^{-3}\text{ mol} \times 8.31\text{ J K}^{-1}\text{mol}^{-1} \times 483\text{ K}}{100\ 000\text{ Pa}}$ = 3.46 x 10^{-4} m^3
  - Convert the volume from m^3 to cm^3
    - 3.46 x 10^{-4} x 1000000 = 346 cm^3

1c

c) To deduce the limiting reagent:

- Moles of P_4 = $\frac{1.85\text{ g}}{(4 \times 31.0)\text{ g mol}^{-1}}$ = 0.0149 (mol)
  AND
  Moles of OH^- = 0.07500 dm^3 x 1.25 mol dm^{-3} = 0.09375 (mol); [1 mark]
- 0.0149 moles of P_4 reacts with 0.0447 moles of OH^-
  OR
  0.09375 moles of OH^- reacts with 0.03125 moles of P_4; [1 mark]
- Phosphorus / P_4 is the limiting reagent; [1 mark]

[Total: 3 marks]

- To determine the limiting reagent, you need to know the number of moles of each reactant involved in the reaction
  - Moles of P_4 = $\frac{\text{mass}}{M_r} = \frac{1.85}{(31.0 \times 4)}$ = 0.0149
  - Moles of OH^- = concentration x volume (in cm^3) $= 1.25 \times \frac{75.0}{1000}$ = 0.09375

- You have to account for the stoichiometric ratio of the reaction
  - 1 mole of P_4 reacts with 3 moles of OH^-, which means that
    - Either 0.0149 moles of P_4 reacts with 3 x 0.0149 = 0.0447 moles of OH^-
    - Or 0.09375 moles of OH^- reacts with $\frac{0.09375}{3}$ = 0.03125 moles of P_4
- Therefore, the P_4 is the limiting reagent

1d

d) The volume of phosphine in cm^3 is:

- 358 (cm^3) to 3 significant figures; [1 mark]

[Total: 1 mark]

- As room temperature and pressure have been stated, you can assume that 1 mole of gas occupies 24.0 dm^3
  - Therefore, the volume of gas at room temperature and pressure = 24.0 x moles
  - This answer will need converting from dm^3 to cm^3, by multiplying by 1000
  - 24.0 dm^3 mol^{-1} x 0.0149 mol x 1000 = 357.6 = 358 cm^3 to 3 significant figures
- You could, alternatively, assume that one mole of gas occupies 24000 cm^3 which removes the need to convert the units from cm^3 to cm^3
  - 24000 cm^3 mol^{-1} x 0.0149 mol = 357.6 = 358 cm^3 to 3 significant figures

2a

a) To calculate the mass of oxygen formed:

Conversion of data to S.I. units:

- P = 303000 (Pa)
  AND
  V = 95.0 x 10^{-6} (m^3)
  AND
  T = 301 (K); [1 mark]

Calculation:

- Moles of O_2 = 1.151 x 10^{-2}

AND

Mass of O_2 = 0.37 g; [1 mark]

[Total: 2 marks]

- You have to convert the data values into S.I. units
  - 303 kPa = 303 x 1000 = 303000 Pa
  - 95.0 cm^3 = 95.0 x 10^{-6} m^3
  - 28 °C = 28 + 273 = 301 K
- The $PV = nRT$ equation needs rearranging for n
  - $n = \frac{PV}{RT}$
- The values are then substituted into the equation, which is evaluated
  - $n = \frac{303000 \times 95 \times 10^{-6}}{8.31 \times 301}$ = 1.151 x 10^{-2} moles of oxygen
- The mass of oxygen is then calculated
  - Mass = moles x M_r = 1.151 x 10^{-2} x (16.0 x 2) = 0.37 g
- Out of good practice, you should be including units in your answers
  - Sometimes your answer will be marked regardless of units and other times it could lose you a mark

2b

b) To calculate the number of electrons involved in the bonding of this sample of oxygen:

- Moles of electrons = 4.604 x 10^{-2}; [1 mark]
- Number of electrons = 2.77 x 10^{22}; [1 mark]

[Total: 2 marks]

- Oxygen molecules contain a double covalent bond, which means that 4 electrons are involved in the bonding
  - i.e. one mole of oxygen has four moles of electrons involved in the bonding
  - Therefore, 1.151 x 10^{-2} moles of oxygen will have 1.151 x 10^{-2} x 4 = 4.604 x 10^{-2} moles of electrons involved in the double covalent bond
- **Remember:** Number of particles = moles x Avogadro's constant
  - Therefore, the number of electrons = 4.604 x 10^{-2} x 6.02 x 10^{23} = 2.77 x 10^{22}

2c

c) O_2 (g) behaves even **less** ideally at very high pressures and very low temperatures because:

Very high pressures:

- The size / volume of molecule / particle becomes significant / non-negligible
  OR
  Intermolecular forces become significant / non-negligible; [1 mark]

Very low temperatures:

- Intermolecular forces become significant / non-negligible
  OR
  Collisions are not elastic; [1 mark]

[Total: 2 marks]

- At very high pressures and low temperatures, real gases do not obey kinetic theory because
  - The molecules / particles are close to each other
  - This means that intermolecular forces (instantaneous dipole-induced dipole or permanent dipole-permanent dipole) become a significant factor pulling the molecules / particles away from the container wall
  - Also, the volume of the molecules is not negligible

2d

d) To write the equation for the complete combustion of **X**:

Conversion of data to S.I. units:

- *P* = 300000 (Pa)
  AND
  $V = 2.0 \times 10^{-3}$ (m^3)
  AND
  T = 293 (K); [1 mark]

Calculation option 1:

- $n = \frac{PV}{RT} = \frac{300000 \times 2.0 \times 10^{-3}}{8.31 \times 293} = 0.2464$; [1 mark]
- $M_r = \frac{\text{mass}}{\text{moles}} = \frac{10.84}{0.2464} = 43.99$; [1 mark]
- Alkane **X** = C_3H_8; [1 mark]

Calculation option 2:

- $M_r = \frac{mRT}{PV}$; [1 mark]
- $M_r = \frac{10.84 \times 8.31 \times 293}{300000 \times 2.0 \times 10^{-3}} = 43.99$; [1 mark]
- Alkane **X** = C_3H_8; [1 mark]

Balanced equation:

- $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$; [1 mark]

[Total: 5 marks]

- To be able to write the equation, you need to know the chemical formula of **X**
  - This can be determined using the information given
- You have to convert the data values into S.I. units
  - 300 kPa = 300 x 1000 = 300000 Pa
  - $2.0\ dm^3 = \frac{2.0}{1000} = 2.0 \times 10^{-3}\ m^3$
  - 20 °C = 20 + 273 = 293 K
- You can either use $PV = nRT$ to calculate the number of moles of **X** and then use moles = $\frac{\text{mass}}{M_r}$ to calculate the M_r of **X**
  - This is shown in "Calculation option 1" of the mark scheme
- Or, you can combine the equations to $PV = \frac{\text{mass} \times RT}{M_r}$ to calculate the M_r of **X**
  - This is shown in "Calculation option 2" of the mark scheme
- Both methods give an M_r of 43.99, which corresponds to propane / C_3H_8
  - **Remember:** The question tells you that **X** is an alkane
- Balancing the equation
  - The question tells you that the reaction is complete combustion
    - $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$
  - The three carbons of propane will form $3CO_2$
  - The eight hydrogens of propane will form $4H_2O$
    - $C_3H_8 + O_2 \rightarrow \mathbf{3}CO_2 + \mathbf{4}H_2O$

- There are now 10 oxygens on the right-hand side, which require $5O_2$ to form
  - $C_3H_8 + \mathbf{5}O_2 \rightarrow 3CO_2 + 4H_2O$

- Or, you can combine the equations to $PV = \frac{}{M_r}$ to calculate the M_r of **X**
  - This is shown in "Calculation option 2" of the mark scheme
- Both methods give an M_r of 43.99, which corresponds to propane / C_3H_8
  - **Remember:** The question tells you that **X** is an alkane
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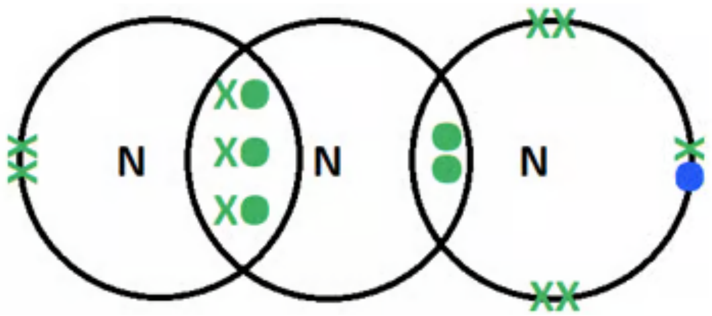
    - $C_3H_8 + O_2 \rightarrow \mathbf{3}CO_2 + \mathbf{4}H_2O$
  - There are now 10 oxygens on the right-hand side, which require $5O_2$ to form
    - $C_3H_8 + \mathbf{5}O_2 \rightarrow 3CO_2 + 4H_2O$

3a

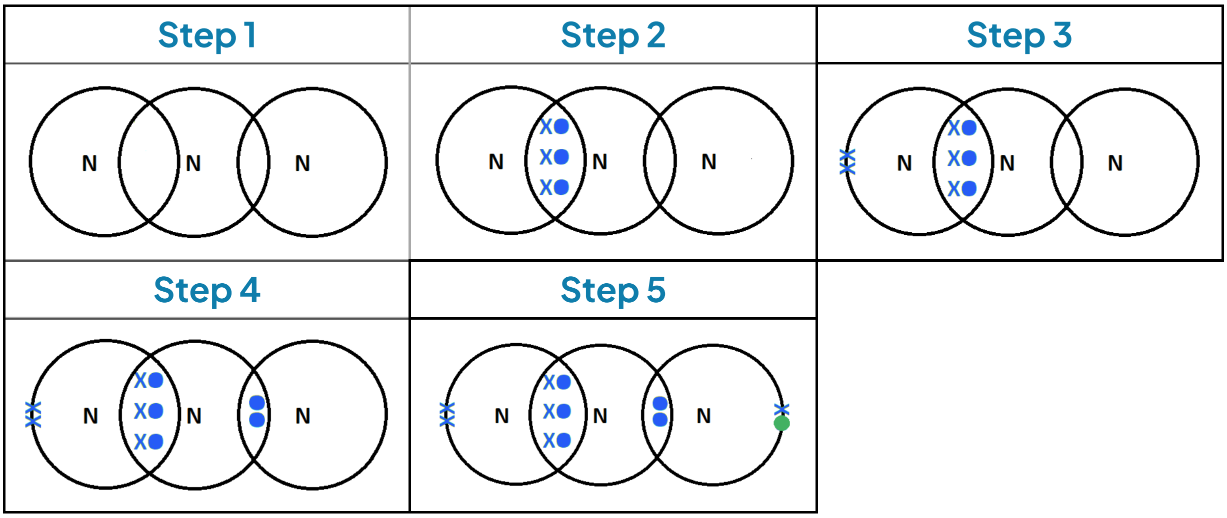
a) The 'dot-and-cross' diagram to show the arrangement of outer electrons present in an azide ion is:



- Central nitrogen atom has 6 shared electrons / 3 shared pairs of electrons
  AND
  2 shared electrons / 1 shared pair of electrons (as a coordinate / dative covalent bond); [1 mark]
- The nitrogen atom with 6 shared electrons / 3 shared pairs of electrons has 2 (non-bonding) electrons / 1 pair of (non-bonding) electrons that are not shared; [1 mark]
- The nitrogen atom with 2 shared electrons / 1 shared pair of electrons (as a coordinate / dative covalent bond) has 6 (non-bonding) electrons
  AND
  One of those electrons must clearly be different, i.e. 5 x and 1 o; [1 mark]

[Total: 3 marks]

- Take your time drawing the azide ion dot-and-cross diagram
- Nitrogen has 5 outer shell electrons
- The azide / N_3^- ion gains one extra electron from the formation of an ionic bond which, in this case, is with sodium
  1. Start the dot and cross diagram by drawing the three nitrogen atoms
  2. Place the 6 electrons / 3 shared pairs of electrons between two of the nitrogen atoms for the triple bond
  3. The left-hand nitrogen atom has 2 electrons left
  4. The central nitrogen atom also has 2 electrons left which are placed in the shared space between the central nitrogen atom and the right-hand nitrogen atom
  5. The right-hand nitrogen atom has 5 electrons plus the extra electron from the 1– charge / ionic bond with sodium which are placed on the right-hand nitrogen **not** in the shared space



3b

b) **Three** properties of sodium azide are:

Any **three** of the following:

- Strong
  AND
  Due to strong electrostatic forces of attraction; [1 mark]
- High melting / boiling point

AND
Due to strong electrostatic forces of attraction; [1 mark]
- Soluble in water
AND
Ion-dipole interactions / bonds can form; [1 mark]
- Conduct electricity when molten / dissolved **AND** ions are mobile / able to move
OR
Does not conduct electricity when solid **AND** there are no ions that are mobile / able to move; [1 mark]

[Total: 3 marks]

- Sodium azide is an ionic compound
- You should know the properties of ionic compounds

3c

c) To calculate the mass, in grams, of sodium amide required to form 550 g of sodium azide with a yield of 95.0%:

- $2NaNH_2 + N_2O \rightarrow NaN_3 + NaOH + NH_3$; [1 mark]
- 100% yield of NaN_3 = 578.95 (g); [1 mark]
- M_r of NaN_3 = 65.0
AND
Moles of NaN_3 = 8.9069; [1 mark]
- Moles of $NaNH_2$ = 17.814
AND
M_r of $NaNH_2$ = 39.0; [1 mark]
- Mass of $NaNH_2$ = 695 g; [1 mark]

[Total: 5 marks]

- This calculation cannot be completed without a correctly balanced equation for the reaction
    - Sodium amide + dinitrogen monoxide → sodium azide + sodium hydroxide + ammonia
        - $NaNH_2 + N_2O \rightarrow NaN_3 + NaOH + NH_3$
    - There are two sodium atoms on the right-hand side which means that $2NaNH_2$ are

required

  - **2**$NaNH_2 + N_2O \rightarrow NaN_3 + NaOH + NH_3$

- To determine the mass of sodium amide required you have to know the mass of sodium azide produced with a 100% yield
  - $\frac{550}{95.0} \times 100 = 578.95$ g
- This allows you to calculate the number of moles of sodium azide, NaN_3
  - M_r of $NaN_3 = 23.0 + (14.0 \times 3) = 65.0$
  - Moles of $NaN_3 = \frac{mass}{M_r} = \frac{578.95}{65.0} = 8.9069$
- Forming one mole of NaN_3 requires 2 moles of $NaNH_2$
  - Moles of $NaNH_2 = 8.9069 \times 2 = 17.814$
- This allows you to calculate the mass moles of sodium amide, $NaNH_2$
  - M_r of $NaNH_2 = 23.0 + 14.0 + (1.0 \times 2) = 39.0$
  - Mass = moles x $M_r = 17.814 \times 39.0 = 695$ g (to 3 significant figures)

3d

d) To calculate the mass of sodium azide that must decompose in order to inflate an airbag to a volume of 7.50×10^{-2} m^3 at a pressure of 150 kPa and temperature of 35 °C:

Conversion of data to S.I. units:

- P = 150000 (Pa)
  AND
  T = 308 (K); [1 mark]

Calculation:

- Moles of N_2 = 4.395; [1 mark]
- Moles of NaN_3 = 2.930; [1 mark]
- Mass of NaN_3 = 190 (g); [1 mark]

[Total: 4 marks]

- For this ideal gas question, you need to:
  - Convert your units for temperature and pressure, make sure you know these from memory:
    - P = needed in Pa (given as kPa in this question so multiply by 1000 to convert to Pa)

150 kPa x 1000 = 150000 Pa

- T = needed in K (given as °C in this question so +273 to convert to K)
  35 ^{o}C + 273 = 308 K

- Rearrange the $PV = nRT$ equation for n
  - $n = \dfrac{PV}{RT}$
- Insert the values and evaluate the expression to get the number of moles of nitrogen
  - $n = \dfrac{150\ 000\ \text{Pa} \times 7.50 \times 10^{-2}\ \text{mol}}{8.31\ \text{J K}^{-1}\text{mol}^{-1} \times 308\ \text{K}} = 4.395$
- Use the molar ratio from the equation to determine the number of moles of sodium azide
  - $2NaN_3$ (s) → 2Na (s) + $3N_2$ (g)
  - 3 moles of nitrogen require 2 moles of sodium azide
  - So, the number of moles of sodium azide = $\dfrac{4.395}{3} \times 2$ = 2.930
- With the moles of sodium azide, you can calculate the mass of sodium azide required
  - Mass = moles x M_r = 2.930 x 65.0 = 190 g (to 3 significant figures)

3e

e)i) The balanced chemical equation for the reaction of sodium azide with acidified nitrous acid is:

$2NaN_3 + 2HNO_2 + 2HCl \rightarrow 3N_2 + 2NO + 2NaCl + 2H_2O$

- $2HNO_2$ + 2HCl; [1 mark]
- $3N_2$ + 2NO + 2NaCl; [1 mark]

ii) To calculate the volume of gas released, in dm^3, when 250 cm^3 volume of nitrous acid is used to destroy 75.0 g of sodium azide:

- Moles of sodium azide = 1.15; [1 mark]
- Moles of nitrogen = 1.73 **AND** moles of nitrogen = 1.15
  OR
  Total moles of gaseous products = 2.88; [1 mark]
- Total volume of gaseous products = Answers in the range of 69.1 to 69.2 dm^3; [1 mark]

[Total: 5 marks]

- For part (i)
  - Starting from the unbalanced equation given
    - $2NaN_3$ + ...HNO_2 + ...HCl → ...N_2 + ...NO + ...NaCl + $2H_2O$
  - $2NaN_3$ must form 2NaCl
    - $2NaN_3$ + ...HNO_2 + ...HCl → ...N_2 + ...NO + **2**NaCl + $2H_2O$
  - 2NaCl require 2HCl
    - $2NaN_3$ + ...HNO_2 + **2**HCl → ...N_2 + ...NO + **2**NaCl + $2H_2O$
  - $2H_2O$ require four hydrogen atoms
    Since there are already two hydrogen atoms from the 2HCl, the other two hydrogen atoms must come from $2HNO_2$
    - $2NaN_3$ + **2**HNO_2 + **2**HCl → ...N_2 + ...NO + **2**NaCl + $2H_2O$
  - $2HNO_2$ contains four oxygen atoms
    Since there are already two oxygen atoms in the $2H_2O$, the other two oxygen atoms must form 2NO
    - $2NaN_3$ + **2**HNO_2 + **2**HCl → ...N_2 + **2**NO + **2**NaCl + $2H_2O$
  - There are eight nitrogen atoms on the left-hand side
    Since there are already two nitrogen atoms in the 2NO, the remaining six nitrogen atoms must form $3N_2$
    - $2NaN_3$ + **2**HNO_2 + **2**HCl → **3**N_2 + **2**NO + **2**NaCl + $2H_2O$
- For part (ii)
  - Calculate the number of moles of sodium azide that is being destroyed in the reaction
    - Moles $= \frac{\text{mass}}{M_r} = \frac{75.0}{65.0} = 1.15$
  - **Careful:** The gaseous products are nitrogen and nitrogen monoxide
  - You have to use the stoichiometric ratio to determine the moles of gas released
    - 2 moles of sodium azide produces 3 moles of nitrogen and 2 moles of nitrogen monoxide
      So, 1.15 moles of sodium azide produces 1.73 moles of nitrogen and 1.15 moles of nitrogen monoxide
      OR
      2 moles of sodium azide produce 5 moles of gaseous products
      So, 1.15 moles of sodium azide produces 2.88 moles of gaseous products
  - **Remember:** One mole of gas occupies 24.0 dm^3 at room temperature and pressure

- 2.88 moles of gaseous product occupies 2.88 x 24.0 = 69.1 dm^3
- Answers in the range of 69.1 to 69.2 are accepted

4a

b)

i) The boiling point of chlorine is different to that of calcium because:

- Chlorine is a simple molecular substance
  AND
  Calcium is metallic **OR** calcium contains positive metal / Ca^{2+} ions surrounded by a 'sea' of delocalised electrons; [1 mark]
- Chlorine has weak van der Waals forces (between molecules) **OR** chlorine has weak intermolecular forces
  AND
  Calcium has strong metallic bonds; [1 mark]
- High(er) / more energy is required to break bonds than overcome forces (between molecules); [1 mark]

ii) Calcium is a liquid over a much greater temperature range compared to chlorine because:

- The forces of attraction in liquid / molten calcium are stronger than in liquid chlorine; [1 mark]

[Total: 4 marks]

- **Remember:** Chlorine is a simple molecular substance
  - Many students forget to mention the structure and lose a mark here
- When chlorine boils, the covalent bonds aren't being broken
  - It is the weak van der Waals forces that are being broken between the molecules of Cl_2
  - **Careful:** Do not say that bonds are being broken here as this is a common misconception

4b

a)

i) The melting points of sulfur dioxide and magnesium oxide are different because:

- Magnesium oxide / MgO is a giant ionic structure
  AND
  Sulfur dioxide / SO_2 is a simple molecular structure; [1 mark]
- The ionic bonds in magnesium oxide / MgO are much stronger than the weak van der Waals forces / intermolecular forces between the sulfur dioxide / SO_2 molecules; [1 mark]
- Therefore, more energy is needed to break (strong) ionic bonds than overcome the weak van der Waals / intermolecular forces
  OR
  Therefore, high(er) / more energy is needed to overcome the (strong) electrostatic forces of attraction than the weak van der Waals / intermolecular forces; [1 mark]

ii) The difference in melting point between sulfur dioxide and ammonia is due to:

- Ammonia / NH_3 has hydrogen bonding between the (ammonia / NH_3) molecules
  AND
  Sulfur dioxide / SO_2 only has weak intermolecular forces / van der Waals forces between the molecules; [1 mark]
- Therefore, high(er) / more energy is needed to overcome the hydrogen bonding in ammonia / NH_3
  AND
  Which means that ammonia / NH_3 has a higher melting point; [1 mark]

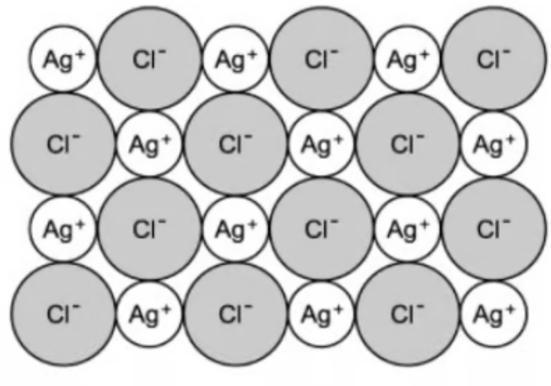
[Total: 5 marks]

- You need to get into the habit of explaining both the structures that are present in species as well as the bonding for these types of questions
  - These two factors must be discussed in order to get full marks
- Compare the strength of bonding / forces and always link this back to the question e.g. melting / boiling points
- Hydrogen bonding exists between hydrogen atoms and electronegative atoms such as fluorine, oxygen and nitrogen
- This is why ammonia has a higher melting point than sulfur dioxide
  - They both have van der Waals forces acting between the molecules
  - But, ammonia also has hydrogen bonding
  - The strength of the hydrogen bonding in ammonia means that more energy is needed to melt ammonia

4c

c)

i) A diagram of the silver chloride lattice showing the arrangement of the ions is:



- 2D arrangement of oppositely charged ions in an ordered lattice arrangement (minimum of 4 ions required); [1 mark]
- Ag^+ in smaller ions and $Cl^–$ in larger ions
  OR
  + charge in smaller ions and – charge in larger ions **AND** labels / key to show that + is Ag^+ and – is $Cl^–$; [1 mark]

ii) The melting point of silver chloride is high because it has:

- There is a large difference in electronegativity between the sodium / Na and chlorine / Cl / Cl_2; [1 mark]
- This indicates there is electron transfer / ions
  OR
  This indicates that there are strong ionic bonds
  OR
  This indicates that there are strong (electrostatic) forces of attraction between oppositely charged particles / ions; [1 mark]
- Which require large amounts of heat energy to overcome; [1 mark]

[Total: 5 marks]

- When you are asked to draw the diagram of an ionic lattice, examiners are often looking for a clear and obvious difference in the size of the ions

    - A variation of this could be to provide the diagram of an ionic lattice and ask for the empirical formula of the ionic compound
    - If this happens, it can be best to ignore the diagram as the empirical formula of an ionic compound is the same as its molecular formula
- For part (ii)
    - The question does not ask you to talk in terms of structure **AND** bonding
        - You should not lose marks for a question like this but it is a better exam technique to always answer this kind of question in terms of structure **AND** bonding
    - The difference in electronegativity can be used to determine if a compound is ionic or covalent
        - Ionic compounds tend to have large differences in electronegativity between the elements involved
        - Covalent compounds tend to have small differences in electronegativity between the elements involved

4d

d) The structure and bonding in diamond and ice are:

- Both diamond and ice have covalent bonding between the atoms; [1 mark]
- Diamond is a giant macromolecular structure; [1 mark]
- All four carbon atoms involved in bonding; [1 mark]
- Diamond has a high melting point as high amounts of energy are required to overcome / break strong covalent bonds; [1 mark]
- Hydrogen bonding between water molecules; [1 mark]
- Covalent bonds between H and O atoms very strong in water; [1 mark]
- Less energy needed to overcome weaker hydrogen bonding in ice, meaning it has a lower melting point (than diamond); [1 mark]

[Total: 7 marks]

- It's important to note that both diamond and ice have covalent bonding in common:
    - It's the arrangement of these atoms in space that creates the different structures
    - Make sure you link the structure and bonding of each substance to the properties that they exhibit in order to get full marks

5a

a) Comparing the electrical conductivity of mercury and mercury(II) bromide:

- Mercury / Hg conducts in both the solid and molten states
  AND
  Mercury(II) bromide / $HgBr_2$ conducts when molten but not when solid; [1 mark]

Explanations:

- The delocalised electrons in mercury move / are mobile and can carry the electrical charge (in both solid and liquid state); [1 mark]
- The ions are mobile in molten $HgBr_2$
  AND
  The ions are fixed in a lattice in solid $HgBr_2$; [1 mark]

[Total: 3 marks]

- This is a comparison question, so take each species at a time to avoid confusing your answer and ensure it is written in a logical order
- **Remember:** Solid ionic compounds cannot conduct electricity as the ions (not electrons!) are fixed within the lattice structure by the strong electrostatic forces of attraction
  - When the ionic compound is molten, the lattice structure has been disrupted and the ions are free to move
- **Remember:** Metals have delocalised electrons which can carry the electrical charge throughout the structure both when molten and in solid states
  - Only the electrons move
  - The positive metal ions (cations) are fixed within the structure

5b

b) The changes in the cation can affect the bond strength in an ionic compound because:

- The higher the charge on the cation, the stronger the attraction / greater the charge density between the ions
  AND
  A higher amount of energy is needed to separate the cation Mg^{2+} in MgF_2 than Li^+ /

Explanations:

- The delocalised electrons in mercury move / are mobile and can carry the electrical charge (in both solid and liquid state); [1 mark]
- The ions are mobile in molten $HgBr_2$
  AND
  The ions are fixed in a lattice in solid $HgBr_2$; [1 mark]

[Total: 3 marks]

- This is a comparison question, so take each species at a time to avoid confusing your answer and ensure it is written in a logical order
- **Remember:** Solid ionic compounds cannot conduct electricity as the ions (not electrons!) are fixed within the lattice structure by the strong electrostatic forces of attraction
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- **Remember:** Metals have delocalised electrons which can carry the electrical charge throughout the structure both when molten and in solid states
  - Only the electrons move
  - The positive metal ions (cations) are fixed within the structure

5b

b) The changes in the cation can affect the bond strength in an ionic compound because:

- The higher the charge on the cation, the stronger the attraction / greater the charge density between the ions
  AND
  A higher amount of energy is needed to separate the cation Mg^{2+} in MgF_2 than Li^+ / Na^+ in LiF / KF; [1 mark]
- The smaller the radius of the cation, the stronger the attraction between the ions
  AND
  The Li^+ ion has a smaller radius and a higher amount of energy needed than the Na^+ ion; [1 mark]

[Total: 2 marks]

- **Careful:** You must use the data from the table in your answer and specifically refer to the ions, not atoms (Li^+, Na^+, Mg^{2+})
- The two factors that affect the energy needed to separate the ions (lattice enthalpy) are:
  - Ionic radius
    - You need to emphasise that a smaller ionic radius means that the centre of the ions will be closer together
    - This results in stronger electrostatic forces of attraction between the ions
    - Therefore, more energy is needed to separate the ions
  - Ionic charge
    - You need to emphasise that a greater charge means that the charge density is higher
    - This results in stronger electrostatic forces of attraction between the ions
    - Therefore, more energy is needed to separate the ions
  - You need to discuss these two factors in your answer but just stating them is not enough to get both marks

5c

b)

i) The type of crystal structure for each of iodine and sodium chloride is:

- Iodine has a simple molecular structure; [1 mark]
- Sodium chloride has a giant ionic lattice structure; [1 mark]

ii) Iodine vaporises easily because:

- It has weak Van der Waals forces / intermolecular forces between the iodine molecules; [1 mark]
- These weak forces of attraction between the molecules are easily broken / overcome, causing iodine to vaporise easily; [1 mark]

iii) The differences in electrical conductivity are explained below:

- Iodine has no free ions to carry the electrical charge; [1 mark]
- Molten sodium chloride has ions which are free to move / mobile and can carry the electrical charge; [1 mark]

- Solid sodium chloride has no free ions as they are held in fixed positions within the lattice and cannot conduct electricity; [1 mark]

[Total: 7 marks]

- **Remember:** The four types of crystalline structures are:
  - Simple covalent / molecular
  - Giant covalent / macromolecular
  - Ionic
  - Metallic
- Iodine / I_2 is a simple molecular structure which has strong covalent bonds between the atoms and weak Van der Waals forces between the molecules:
  - You must refer to the weak Van der Waals forces being overcome between the molecules not atoms when explaining why it vaporises easily
- **Remember:** Iodine has no charged species, as it is a simple molecular substance
- Sodium chloride can only conduct electricity when molten /dissolved, as the ions are mobile
  - Otherwise, it cannot conduct electricity because the ions are in fixed positions in the lattice structure when solid

5d

d) Graphene and graphite are good conductors of electricity yet diamond is not because:

- Graphene has a single layer / sheet of carbon atoms / rings / hexagons; [1 mark]
- Graphene has delocalised electrons / free moving/mobile electrons which can carry the electrical charge; [1 mark]
- Graphite has multiple layers / sheets / planes of carbon
  AND
  Each carbon atom is bonded (covalently) to three other carbon atoms; [1 mark]
- Graphite has delocalised electrons / free moving / mobile electrons which can carry the electrical charge; [1 mark]
- Diamond has a tetrahedral structure of carbon atoms
  AND
  Each carbon atom is bonded (covalently) to four other carbon atoms; [1 mark]
- All four outer shell electrons are involved in bonding meaning there are no delocalised electrons / free moving / mobile electrons which can carry the electrical charge; [1